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AI-Assisted Criminal Investigations: Enhancing Testimony Analysis and Case Correlation

Dynamic Question Generation and Response Analysis

R25-020

Final Report

IT21822230

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Declaration

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books)

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The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Signature of the supervisor: Date:

Acknowledgement

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Abstract

This research addresses critical limitations in traditional criminal investigations by developing an AI-assisted system that enhances testimony analysis and dynamic interrogation processes. Conventional interrogation methods rely extensively on investigator expertise, which can be affected by cognitive biases, fatigue, and subjective judgment, leading to inefficiencies and compromised accuracy. Leveraging advances in Natural Language Processing (NLP) and transformer-based Large Language Models (LLMs), this system introduces dynamic question generation tailored in real-time to suspect responses. It integrates real-time analysis of testimony to detect inconsistencies, contradictions, and missing information, thereby providing investigators with actionable and contextually relevant follow-up questions. The system further incorporates multimodal behavioral analytics including emotion and stress indicators from speech and facial cues to enrich interrogation insights. Architected as a scalable full-stack application with React/Next.js frontend, MongoDB backend, and APIs interfacing with LLMs, the platform supports secure, multi-session interrogations with an intuitive investigator interface. Comprehensive feasibility studies, field visits, and data collection from Sri Lankan law enforcement ensured alignment with operational, legal, and ethical requirements. Experimental evaluation demonstrated the LLM's superior semantic relevance for question generation against expert-curated benchmarks. The system maintains ethical safeguards such as human oversight, data anonymization, and bias monitoring to ensure fairness and judicial compliance. This AI-assisted interrogation assistant promises to enhance investigation efficiency, accuracy, and ethical standards while accelerating case resolution and strengthening the criminal justice process.

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1. Introduction

1.1.Background & Literature Survey

Criminal investigations rely heavily on the quality and reliability of witness testimonies and suspect interrogations. Traditionally, these interrogations depend on the investigator's expertise to formulate questions, interpret responses, and identify inconsistencies. However, this manual approach is inherently limited by human factors such as cognitive bias, fatigue, and subjective judgment. Furthermore, documentation processes for interrogations often involve manual note-taking or audio/video recordings that require time-consuming post-interview analysis, increasing the risk of errors and missed details. These limitations affect the efficiency and accuracy of investigations, prompting the need for technological advancements that can augment the investigative process.

Recent advancements in Artificial Intelligence (AI), particularly in Natural Language Processing (NLP), have paved the way for automated systems that can analyze textual data with greater nuance and contextual understanding. Transformer-based architectures like BERT and GPT have revolutionized the capability of machines to understand and generate human language effectively. Their applications span diverse fields, including customer service, education, and legal domains. However, their full potential remains untapped in criminal investigations where questioning must be adaptive, context-aware, and responsive to dynamic interactions during interrogations.

Dynamic question generation represents a promising innovation by enabling systems to generate follow-up questions tailored to suspect responses in real-time. Unlike static, pre-scripted interrogation methods, dynamic questioning adapts to evolving narratives and behavioral cues, thereby enhancing the possibility of uncovering contradictions and hidden information. Integrating behavioral and emotional analysis further enriches this process by detecting stress, hesitation, or deception through vocal tone and facial expressions. Such multi-modal integration supports a more comprehensive evaluation of testimony credibility and case evidence.

The proposed system aims to address existing gaps by leveraging state-of-the-art NLP models and real-time response analysis, coupled with behavioral analytics, to assist investigators during interrogations. This AI-assisted approach is designed to reduce human error, minimize investigator fatigue, and provide objective insights that enhance the depth and accuracy of criminal testimony analysis, ultimately contributing to faster and more reliable case resolutions.

Recent research highlights the growing role of Artificial Intelligence (AI) in enhancing criminal investigations, particularly through Natural Language Processing (NLP) and machine learning. Das et al. emphasized that AI-driven frameworks can significantly improve forensic efficiency by automating evidence interpretation and supporting investigators with advanced analytical tools [1]. Similarly, Sabherwal and Kaur argued that AI has the potential to improve the fairness and consistency of criminal justice systems by reducing cognitive bias and human error during investigations [2].

Advances in text classification and contextual understanding provide a strong foundation for dynamic question generation. Hu et al. demonstrated the effectiveness of Convolutional Neural Networks (CNNs) in short text classification, improving semantic analysis accuracy for real-time applications [3]. Lai et al. further advanced this domain through recurrent convolutional neural networks that capture sequential dependencies in responses, which is vital for detecting inconsistencies in testimonies [4]. Conneau et al. extended these efforts by applying very deep convolutional networks for context-sensitive text understanding, enabling models to handle complex interrogation narratives [5].

Together, these studies underscore the potential of LLMs and deep learning techniques in automating testimony analysis. However, their application in adaptive questioning within high-stakes environments like interrogations remains underexplored, leaving a critical gap that this research seeks to address.

1.2.Research Gap

Despite the growing interest and advancements in applying Artificial Intelligence (AI) to criminal investigations, particularly in the domain of Natural Language Processing (NLP) and machine learning, significant gaps remain that hinder the full realization of AI's potential to enhance interrogation processes. Traditional criminal investigations heavily rely on manual interrogation methods, which are limited by human cognitive biases, fatigue, and subjective judgment. Existing AI applications primarily focus on isolated tasks such as lie detection, evidence interpretation, or basic textual analysis but lack integrated solutions that dynamically adapt questioning in real-time based on suspect responses.

Current NLP models and large language models (LLMs) like Gemini and GPT have demonstrated excellent capabilities in text understanding and contextual analysis across various domains but have not been sufficiently tailored or leveraged for the specific and high-stakes environment of criminal interrogations. Most existing research emphasizes static or pre-scripted questioning strategies rather than dynamic, context-sensitive follow-ups that can reveal contradictions and deeper insights during live interrogations.

Moreover, integration of multi-modal behavioral analytics, such as emotional and stress detection through voice tone or facial expressions, is not widely explored or incorporated into AI-assisted interrogation frameworks.

Furthermore, the practical application of these advanced technologies is constrained by the challenges faced in real-world law enforcement workflows, including data privacy, ethical considerations, investigator usability, and seamless integration with existing case management systems. There is a noticeable lack of comprehensive AI-driven investigative platforms that combine dynamic question generation, real-time response analysis, and behavioral insights into a user-friendly interface designed specifically for investigative personnel.

Therefore, this research addresses a critical gap by designing and developing an AI system that not only leverages state-of-the-art NLP and machine learning technologies for adaptive questioning and response analysis but also integrates behavioral analytics to support investigators throughout the interrogation process. By doing so, it aims to reduce human error, cognitive biases, and fatigue while enhancing the accuracy, efficiency, and ethical standards of criminal investigations.

1.3.Research Problem

Traditional criminal investigation methods rely heavily on manual interrogation techniques where investigators formulate questions, interpret suspect responses, and identify inconsistencies based primarily on their expertise and judgment. This manual approach is prone to significant limitations such as cognitive biases, human fatigue, and subjective interpretation, which can compromise the accuracy, consistency, and depth of testimony analysis. Moreover, the documentation and analysis of interrogations involve time-consuming manual processes that increase the risk of overlooking critical details and hinder timely case resolution.

Despite advances in Artificial Intelligence (AI) and Natural Language Processing (NLP), current investigative tools are generally narrow in scope ,focusing on isolated tasks such as lie detection, basic evidence interpretation, or static questioning strategies without real-time responsiveness. There is a lack of integrated platforms that can dynamically generate contextually relevant follow-up questions tailored to suspect responses during live interrogations, together with real-time analysis to detect contradictions and semantic inconsistencies. Additionally, the incorporation of multi-modal behavioral analysis, such as emotional and stress detection through voice and facial cues, remains underutilized in supporting interrogation effectiveness.

Furthermore, existing AI applications for criminal investigations face challenges in practical deployment, including the need for user-friendly interfaces suited to investigators' workflows, safeguarding data privacy and ethical standards, and seamless integration with existing law enforcement case management systems. These gaps lead to inefficiencies, errors, and lost investigative opportunities that delay case closures and potentially impact judicial outcomes.

This research problem centers on developing a comprehensive AI-driven system that addresses these shortcomings by leveraging state-of-the-art NLP and machine learning models to provide dynamic, adaptive questioning and robust response analysis. The system must also integrate behavioral analytics and enable investigator interaction through an intuitive interface, thereby reducing human error, mitigating cognitive bias, and improving the overall efficiency, accuracy, and ethical integrity of criminal investigations.

1.4. Research Objectives

Main Objective

The main objective of this project is to design, develop, and evaluate an advanced AI-driven system that integrates state-of-the-art Natural Language Processing (NLP) and machine learning techniques to dynamically generate contextually relevant follow-up questions and conduct real-time response analysis during criminal interrogations. This system aims to transform traditional interrogation methodologies by enabling adaptive, evidence-based questioning that responds intelligently to suspect replies, thereby enhancing the accuracy and depth of testimony evaluation. Key functionalities include the detection of inconsistencies and contradictions within suspect statements. By doing so, the system seeks to mitigate the adverse effects of investigator cognitive biases and fatigue, which commonly hinder the effectiveness of human interrogators.

Furthermore, the research endeavors to create an intuitive investigator interface that facilitates seamless interaction with AI-generated insights, allowing for question approval and refinement, as well as incorporating real-time feedback to continuously improve the system's performance. The module is intended to integrate securely with law enforcement workflows, maintaining data privacy and complying with relevant legal and ethical standards. Ultimately, this dissertation aims to provide criminal justice agencies with a powerful, interactive tool that not only improves the efficiency and consistency of interrogations but also accelerates case resolution by uncovering critical evidence links and strengthening the reliability of collected testimonies. Through this research, the promise of AI-assisted investigations will be advanced, contributing to more ethical, accurate, and expedited criminal justice outcomes.

Specific Objectives

The following specific objectives outline the core tasks necessary to achieve the main aim of this research. They focus on building an AI-driven system capable of generating dynamic, context-aware questions, analyzing responses, and supporting investigators through an intuitive and secure platform.

- **Develop an LLM-powered module for dynamic question generation**
  Create a system that utilizes Large Language Models to generate contextually relevant and adaptive follow-up questions based on suspect responses and predefined interrogation parameters.

- **Implement an LLM-based response analysis engine**
  Design a model capable of detecting inconsistencies, contradictions, and recurring patterns in suspect testimonies, thereby enhancing the accuracy and depth of interrogation outcomes.

- **Design a user-friendly investigator interface**
  Build an intuitive interface that allows investigators to seamlessly interact with the system, review AI-generated questions, and provide real-time feedback on their effectiveness.

- **Establish a continuous feedback mechanism**
  Integrate a feedback loop where investigator evaluations are used to refine the model, improving the contextual accuracy, adaptability, and overall quality of generated questions.

- **Develop a secure and scalable backend system**
  Ensure robust backend infrastructure for data storage, model integration, and secure handling of interrogation data, while maintaining compliance with privacy and ethical standards.

2. Methodology

2.1. Requirements Gathering

The requirement gathering and analysis phase formed the foundation of this research, ensuring that the system design was guided by the actual needs of investigators in Sri Lanka. This stage was critical in aligning the technological solution with the practical challenges faced by law enforcement personnel in handling testimonies, case reports, and databases. The process combined multiple approaches, including a literature survey, field visits, interviews, and data collection. These methods provided both theoretical insights and empirical evidence to identify the functional and non-functional requirements of the proposed system.

Literature Survey

The requirements gathering process commenced with a detailed literature survey that reviewed existing studies on AI applications in criminal investigations, dynamic question generation, and testimony analysis. Prior work on Natural Language Processing (NLP), Large Language Models (LLMs), deception detection, and semantic analysis was examined to identify state-of-the-art methods and their limitations in interrogation contexts. This step provided the theoretical basis for designing the system architecture and highlighted gaps in adaptability, and real-time analysis.

Field Visit

To validate the findings of the literature review and gather firsthand insights, field visits were conducted at two police stations in Sri Lanka. These visits were instrumental in understanding the day-to-day operations of investigators and the administrative processes associated with managing case data.

- Interviews with police officers to explore the practical difficulties encountered during investigations, such as the time required to manually review case files, challenges in cross-referencing testimonies, and issues in retrieving archived reports
- Observing real case reports and files to examine the structure, format, and language used in documentation. This provided direct exposure to the nature of unstructured data, highlighting the variations in style, terminology, and completeness of reports.
- Conducted further online interviews with police personnel to understand the different procedures followed and challenges faced during the investigation process.

- Sought frequent guidance from a retired police officer as our external supervisor to ensure our system design aligns with real investigative practices.

Data Collection

Following the field visits, anonymized police reports from past incidents were collected to create a dataset for experimentation and model training. These reports were carefully curated to ensure compliance with privacy and ethical standards, with all personally identifiable information removed. Additionally, sample police reports were created to simulate real investigative scenarios, ensuring that the system could be tested on both authentic and synthetic data.

The data collection process served multiple purposes. Firstly, it provided the training material for NLP and LLM-based modules responsible for entity recognition, semantic classification, and testimony correlation. Secondly, it enabled the development of preprocessing pipelines for cleaning, anonymizing, and standardizing data prior to analysis. Finally, the collected reports acted as test cases during the pilot evaluation phase, ensuring that the system's accuracy, scalability, and usability were assessed against real-world scenarios.

2.2.Feasibility Study

The feasibility of developing an AI-assisted system for enhancing testimony analysis and case correlation in criminal investigations has been carefully evaluated from multiple perspectives including technical, operational, legal, and economic considerations.

- **Technical Feasibility**

The technical feasibility of this project is grounded in the rapid advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP), particularly with the emergence of transformer-based Large Language Models (LLMs) such as OpenAI's GPT family and Google's Gemini. These models have demonstrated exceptional capabilities in natural language understanding, semantic reasoning, and dynamic text generation, making them ideal for interrogation contexts where adaptive questioning and nuanced response analysis are required. By fine-tuning these LLMs on domain-specific datasets such as anonymized police reports, the system can be tailored to detect inconsistencies, identify contradictions, and generate contextually relevant follow-up questions in real time. Unlike rule-based or static systems, transformer-based architectures excel at adapting to evolving dialogues, a feature critical for interrogations where testimonies shift dynamically during questioning.

The availability of anonymized police reports and synthetic datasets, which provide the necessary resources for model training, validation, and evaluation. Real-world police reports offer insights into the structure, terminology, and linguistic patterns common in testimonies, while synthetic datasets allow controlled testing of edge cases and rare scenarios. To ensure compliance with ethical and legal standards, the documents will be preprocessed for data anonymization, cleaning, and standardization will be implemented, eliminating personally identifiable information (PII) while retaining the contextual richness required for training AI models. This approach ensures that the datasets used in model development adhere to privacy requirements while also maintaining consistency and quality across different input sources.

On the infrastructure side, the proposed system architecture is designed to be scalable, modular, and secure. A microservices-based backend will allow different components to operate independently while seamlessly integrating into a unified system. For data management, NoSQL relational databases such as MongoDB will provide secure storage of interrogation logs, feedback, and analysis results, with built-in mechanisms for access control.

In summary, the project is technically feasible due to the maturity of LLMs, the availability of multi-modal AI tools, and the existence of robust infrastructure frameworks for deployment and integration. By leveraging these advancements, the system can deliver a reliable, adaptive, and ethically compliant interrogation assistant capable of supporting investigators in real-world scenarios. The combination of state-of-the-art AI, carefully curated datasets, and a secure, scalable architecture ensures that the technical foundation of the project is both solid and future-proof.

The operational feasibility of this project has been carefully assessed through direct engagement with law enforcement personnel and the observation of day-to-day investigative practices. Field visits to local police stations, combined with structured interviews and follow-up discussions with officers, provided first-hand insight into the limitations of existing interrogation processes. These interactions highlighted the time-consuming nature of manual testimony review, the difficulty in cross-referencing large volumes of unstructured reports, and the challenges associated with identifying contradictions and behavioral cues during interrogations. Such evidence validated the operational need for an AI-assisted solution capable of supporting investigators in real time. By ensuring that the proposed system is designed in alignment with these identified needs, the likelihood of practical adoption and sustained use is significantly increased.

- **Operational Feasibility**

A major factor in operational feasibility is the development of a user-friendly investigator interface that facilitates seamless interaction with the system. The proposed interface is designed to present AI-generated insights ,such as follow-up question suggestions, detected inconsistencies, and emotional cues ,in a clear and interpretable format. Real-time interaction and feedback mechanisms enable investigators to approve, modify, or reject system outputs during interrogations, ensuring that the technology supports rather than overrides human decision-making. This interactive design not only promotes ease of adoption but also encourages trust in the system by allowing investigators to remain in full control of interrogation outcomes.

Furthermore, security and ethical considerations have been integrated into the operational design of the system to ensure compliance with both legal and organizational standards. All interrogation data will be anonymized, securely stored, and accessed only through authorized credentials, minimizing risks of misuse or data breaches. Ethical safeguards ,such as transparency in AI decision-making, accountability through audit trails, and the prevention of coercive questioning patterns ,have been incorporated to ensure that the system remains compliant with Sri Lankan legal frameworks and international best practices.

The involvement of law enforcement personnel as an external supervisor further strengthens operational feasibility by ensuring that system functionalities reflect real investigative practices rather than abstract technological assumptions. His continuous feedback has helped bridge the gap between technical capabilities and procedural realities, ensuring that the system not only addresses operational challenges but also adheres to investigatory norms. This alignment with practical workflows, combined with a strong emphasis on usability, security, and ethics, positions the proposed system as a viable and sustainable tool within law enforcement operations.

- **Legal and Ethical Feasibility**

The handling of sensitive investigative data within this project necessitates strict compliance with legal regulations, data privacy standards, and ethical guidelines. Criminal investigations involve testimonies, case reports, and other records that often contain personally identifiable information. To address this, the system incorporates anonymization techniques to remove or mask sensitive details during data preprocessing, ensuring that all data used for model training and evaluation is ethically compliant and legally admissible. Additionally, the backend architecture includes secure storage and controlled access protocols, with encryption and authentication layers designed to prevent unauthorized use or breaches. These mechanisms are aligned with international standards of data protection and are adaptable to emerging local legal frameworks in Sri Lanka.

A crucial aspect of legal feasibility lies in the alignment with judicial requirements and evidentiary standards. For AI-assisted tools to be operational in law enforcement and potentially admissible in court proceedings, outputs must be explainable, auditable, and free from coercive influence. The system addresses this by incorporating audit trails for every generated question and response analysis, ensuring that investigators and courts can review the decision-making process. This transparency not only strengthens the credibility of the system's outputs but also safeguards against legal challenges related to bias, reliability, or procedural fairness.

From an ethical standpoint, the system is designed to support investigators rather than replace them, ensuring human oversight at every stage of interrogation. The AI-generated follow-up questions and response analyses are provided as recommendations, leaving final decisions to investigators. This mitigates risks of overreliance on technology and helps preserve human judgment and accountability in sensitive interrogation contexts. Additionally, ethical considerations extend to bias detection and mitigation within LLMs, as language models trained on unbalanced datasets risk perpetuating unfair stereotypes. To counter this, the system integrates continuous monitoring and model refinement mechanisms based on investigator feedback and real-world validation.

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The project also remains informed by ongoing regulatory developments regarding the use of AI in law enforcement. Given the global discourse on AI ethics, fairness, and transparency, the system is designed with the flexibility to adapt to evolving legal standards and policy frameworks. Ethical safeguards such as non-coercive questioning patterns, preservation of suspect rights, and data confidentiality have been explicitly embedded into the design. By combining technical privacy safeguards, legal compliance, and a clearly defined ethical framework, the system ensures that its outputs can be trusted not only by investigators but also by judicial authorities and the wider public.

2.3.System Design

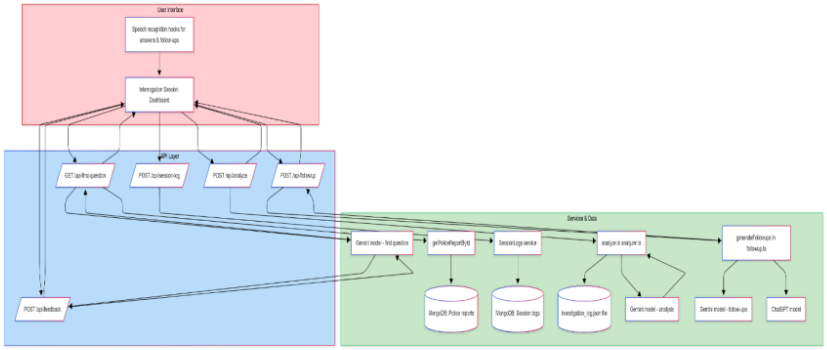

Figure 1: System Overview Diagram

The dynamic question-generation and response analysis system comprises a modern full-stack application built around a React/Next.js front-end, a set of server-side API endpoints, a persistence layer backed by MongoDB, and integrations with large language models such as Google's Gemini and OpenAI's GPT-4. The following section explains the major components and their interactions, justifying design decisions with references to external sources.

1. **Frontend (Interrogation Session Dashboard)**

The investigator interface is a Next.js client component that presents the interrogation dashboard. It displays the suspect's case details, controls the session lifecycle (start, pause/resume, stop), shows a live timer, and renders prompts, answers and analytical feedback. State management is handled with React hooks and the UI is styled with Tailwind CSS components. Voice input is enabled via a custom useSpeechRecognition hook, which wraps the browser's Web Speech API. This abstraction allows investigators to record spoken answers and manually dictate follow-up questions, improving ease of use in field settings.

All network interactions occur through fetch calls to api endpoints. The front-end never directly communicates with the database or LLMs; instead, it relies on typed JSON responses returned by the server. This separation of concerns keeps the client lightweight and secure.

2. **API layer (Next.js API routes)**

- **first-question (GET)** – Generates the initial open-ended question. The route uses getPoliceReportById() to fetch the case record from MongoDB and formats a detailed prompt describing the incident, victim, perpetrator and other contextual fields. It then calls the Gemini API via Google's @google/generative-ai client with the model gemini-2.0-flash. The Gemini API accepts a model name and contents array and returns text output, which the handler extracts and sends back to the front-end as the initial question.

- **analyze (POST)** – Analyses the suspect's answer. The handler invokes the analyze() function. This function constructs a structured prompt instructing the model to rate relevance (0–10), list facts gathered, highlight missing information and suggest points for deeper probing. The same Gemini model is called to generate the analysis. The raw LLM output is parsed into structured fields (relevanceScore, dataGathered, missingData and pointsToExplore) and appended to an investigation_log.json file for auditing.

- **followup (POST)** – Generates follow-up questions. The service calls two generative models: Google's Gemini and OpenAI's GPT-4. The Gemini call uses the same client library as above, while the ChatGPT call uses the official openai Node SDK. As shown in a tutorial on building a CLI chatbot with the ChatGPT API, the OpenAI client is created with an API key and the method is invoked with a list of messages (system prompt and user prompt) and the desired model. The follow-up handler returns one question from each model and indicates the preferred model, allowing the investigator to choose which LLM to trust in subsequent steps.

- **feedback (POST)** – Accepts the investigator's selection (either a generated follow-up or a manually typed question) and returns the next primary question.

- **session-log (POST)** – Persists the session. Before concluding the session, the client sends an array of entries (questionId, question, answer and numeric relevance score) along with a sessionId and investigationId. This endpoint interacts with the SessionModel through Mongoose. The service uses findOneAndUpdate() with upsert: true to perform atomic updates: if a session log for the given sessionId exists, it updates the document; otherwise it inserts a new document. This ensures that concurrent updates to the same session are consistent, and no duplicates occur.

3. **Services and persistence layer**

Police report service

The function abstracts database access for retrieving case summaries. It uses Mongoose models to query the PoliceReports collection in MongoDB. By encapsulating data retrieval in a service, the API handler remains focused on prompt construction and result formatting.

Session log service

The session log service wraps Mongoose operations for creating, updating, querying and deleting Session documents. It uses ensureObjectId() to validate that incoming IDs are valid MongoDB ObjectIds. At the end of a session, the service writes the session log to the database along with timestamps and optional caseIds. This persistent record supports auditing and analysis of interrogation strategies.

In-memory and file logging

Besides database storage, the analyzer module writes each analysis entry as a JSON line to database. This file serves as an immediate audit trail and can be imported into analytics tools for offline review.

LLM integration

The core intelligence of the system relies on two families of large language models:

1. **Google Gemini (Generative AI)** – The Gemini API supports multi-modal input but is used here purely for text generation. The system leverages this for first-question generation, answer analysis and follow-up question generation. Using a consistent model ensures that the follow-up questions remain aligned with the initial interviewing style.
2. **OpenAI GPT-4 via Chat Completions** – For alternative follow-ups, the system uses OpenAI's GPT-4 through the Node SDK. The message structure includes roles for "system" instructions, "user" questions and "assistant" responses, enabling context-aware conversation. This design allows the investigator to

compare the questioning styles of two different LLMs and select the most appropriate follow-up.

Architectural rationale

This architecture follows the principles of separation of concerns and scalability. By using Next.js API routes, the project can be deployed as a single serverless application where API handlers and UI coexist but operate in different execution contexts. Ensuring that sensitive LLM calls and database operations never run on the client. Integrating two different LLM providers allows for richer questioning strategies and reduces model bias. Logging analyses both to a database and a file provides redundancy and facilitates offline investigation. Finally, using a speech recognition hook allows the application to support voice-first workflows.

2.4. Implementation

The proposed system has been designed to replicate the natural flow of an interrogation session while embedding automation, intelligence, and adaptive questioning capabilities. Each session progresses through a series of structured stages, ensuring seamless interaction between the investigator and the AI system. The workflow is as follows:

```
tend > app > api > first-question > [...query] > TS route.ts > ...
  export async function GET(
      );
    }
    const genAI = new GoogleGenerativeAI(apiKey);
    const model = genAI.getGenerativeModel({ model: "gemini-2.0-flash" });
    const prompt =
      `You are a professional criminal investigator.\n\n` +
      `Below is a police case summary extracted from an incident report:\n\n` +
      `Case ID: ${caseData.caseId || ""}\n` +
      `Incident Date & Time: ${caseData.incidentDate || ""} at ${
        caseData.incidentTime || ""
      }\n` +
      `Report Date: ${caseData.reportDate || ""}\n` +
      `Victim: ${caseData.victimName || ""}, Age: ${
        caseData.victimAge || ""
      }, Gender: ${caseData.victimGender || ""}\n` +
      `Perpetrator: ${caseData.perpetratorName || ""}\n` +
      `Relationship to Victim: ${caseData.relationshipToVictim || ""}\n` +
      `Location: ${caseData.incidentLocation || ""}\n` +
      `Summary: ${caseData.incidentSummary || ""}\n` +
      `Type of Violence: ${caseData.typeOfViolence || ""}\n` +
      `Injuries: ${caseData.injuryDescription || ""}\n` +
      `Evidence: ${caseData.evidenceMentioned || ""}\n` +
      `Action Taken: ${caseData.actionTaken || ""}\n` +
      `Recurrence: ${caseData.recurrence || ""}\n` +
      `Case Status: ${caseData.caseStatus || ""}\n` +
      `Relevant Laws: ${caseData.relevantLaws || ""}\n` +
      `Prior Criminal History: ${caseData.priorCriminalHistory || ""}\n\n` +
      `Based on the details above, generate the first open-ended question an investigator should ask the victim to begin the interview. Re
```

Figure 2 : Data Extraction

1. **Start Session**
   The interrogation process begins when the investigator initiates the session by clicking the "Start Session" button on the interface. This action triggers a request to the first-question API endpoint, passing the unique case ID and the role of the individual under interrogation (victim or perpetrator). The API handler retrieves the corresponding police report from the database and constructs a rich context prompt, embedding details such as incident summaries, entities, and prior testimonies. This context is then sent to Gemini, which generates an opening question tailored to the case background. The resulting question is immediately displayed on the investigator's interface, ensuring the session begins with a focused, evidence-driven prompt. Capture answer – The suspect responds verbally. The useSpeechRecognition hook transcribes the speech and populates the answer state. React state updates propagate through the UI.

2. **Capture Answer**
   Once the suspect responds, the system employs the useSpeechRecognition hook to capture the verbal response in real time. The audio is transcribed into text and stored within the React state, which continuously updates the user interface to display the suspect's answer. This transcription serves two purposes:

   i. maintaining an immediate record of the suspect's statement, and
   ii. preparing structured input for subsequent AI analysis. This feature significantly reduces reliance on manual notetaking, ensuring accuracy and minimizing transcription delays during the interrogation

3. **Analyze Answer**
   At this stage, the investigator triggers an analysis of the captured response by calling the analyze API endpoint. The current question–answer pair is transmitted to the backend server, which then queries Gemini. The model evaluates the answer for relevance, coherence, and completeness, while also extracting key data points (e.g., names, dates, locations, and described events). Additionally, the analysis highlights missing or ambiguous information and suggests lines of inquiry for further exploration. The structured output is then returned to the dashboard, where it is visualized by the investigator.

4. **Generate Follow-ups**
   To deepen the interrogation, the investigator clicks the "Generate Follow-ups"

button. This action sends the Q&A pair to the followup API endpoint, which queries both LLMs in parallel. Each model generates one candidate follow-up question, resulting in at least two distinct suggestions. Alongside the questions, the system provides an indication of which model's response is likely more contextually relevant, based on internal evaluation metrics. The investigator is then given flexibility: they may select one of the suggested follow-ups or alternatively formulate a manual question. This design ensures that AI recommendations assist investigators without removing human control from the interrogation process.

5. **Submit Feedback**
   Once a follow-up question is chosen, it is sent to the feedback endpoint. The backend processes the selection, updating the reinforcement component of the system. Positive selections reinforce the generation patterns of the chosen model, while manual overrides provide feedback on model limitations. The selected question is also used by the corresponding LLM to formulate the next primary interrogation prompt, effectively closing the loop and preparing the system for the subsequent cycle of questioning. This continuous loop establishes a reinforcement-based improvement mechanism, where investigators input directly refines future question generation.
6. **Stop Session**
   When the interrogation concludes, the investigator terminates the session by clicking the "Stop Session" button. The client calls the session-log APIendpoint, sending the complete record of session entries. The server commits this data to MongoDB using the findOneAndUpdate() method with the upsert option, ensuring atomic updates that either create a new record or update an existing one. This guarantees that the session log is securely stored, audit-ready, and retrievable for future reference, case correlation, or judicial review.

2.5. Technology Used

- **Next.js and React**
  React is a popular JavaScript library for building user interfaces, while Next.js is a React-based framework that provides server-side rendering, static site generation, and an API routing mechanism. The project's frontend is built with React for its component-based architecture and responsiveness, while Next.js powers the backend API routes creating a unified full-stack framework. This eliminates the need for separate frontend and backend servers.
  Justification: Next.js was chosen over alternatives such as Create React App or Angular because of its integrated API routes, SEO benefits, and scalability. Its

hybrid rendering (static + dynamic) makes it more efficient for real-time interrogation sessions compared to single-page frameworks.

- **TypeScript**
  TypeScript is a superset of JavaScript that adds static typing, improving code quality and reducing runtime errors. Used across the frontend and backend, TypeScript enforces type safety for API calls, state management, and MongoDB schema interactions.
  Justification: TypeScript was preferred over plain JavaScript because it reduces bugs in a mission-critical project, ensures consistency in large codebases, and improves developer productivity by enabling better IntelliSense and debugging in Visual Studio Code.
- **Tailwind CSS**
  Tailwind CSS is a utility-first CSS framework that provides low-level classes to design modern interfaces without writing custom CSS from scratch. Tailwind is used for styling the dashboard, interrogation interface, and forms, ensuring a clean, responsive, and professional look.
  Justification: Compared to Bootstrap or Material UI, Tailwind was chosen for its flexibility, minimalism, and ease of customization, allowing the project to maintain a unique but consistent UI without being constrained to prebuilt styles.

- **ShadCN/UI Components**
  ShadCN is a UI component library that offers prebuilt, customizable React components like cards, buttons, tabs, and separators. Provides a consistent look and feel across the application, speeding up UI development while maintaining high design quality.
  Justification: ShadCN was chosen over libraries like Chakra UI or Ant Design because of its close integration with Tailwind CSS, enabling seamless design consistency while keeping the project lightweight.
- **Lucide icons**
  Lucide is an open-source icon library for modern applications. Provides intuitive icons for UI elements such as buttons, navigation, alerts, and indicators, improving usability.
  Justification: Chosen over Font Awesome or Heroicons due to its lightweight design, extensive library, and compatibility with ShadCN and Tailwind, ensuring a cohesive visual experience.
- **MongoDB**

MongoDB is a NoSQL, document-oriented database. Used to manage structured interrogation data, including session logs, questions, answers, feedback, and cross-referenced case reports.
Justification: MongoDB was chosen over SQL-based alternatives like PostgreSQL or MySQL because of its flexibility in handling unstructured and semi-structured data, scalability, and seamless integration with Next.js APIs.

- **Google Cloud Console (APIs)**

  Google Cloud provides a suite of APIs and cloud services for hosting, training, and deploying machine learning models. Enables fine-tuning of LLMs and secure integration of Google's Generative AI models with the system.
  Justification: Google Cloud was chosen over AWS and Azure because of its native support for Gemini APIs and Generative AI SDK, making it the most direct and efficient choice for experimentation with Google's LLMs.

- **Google Generative AI SDK - LLM fine tuning**

- **Gemini**

  Google's state-of-the-art Large Language Model optimized for text understanding and generation.
  Serves as the primary engine for generating the first question, analyzing suspect answers, and suggesting follow-up questions.
  Justification: Chosen for its strong contextual reasoning and efficiency, Gemini was found to outperform other models like Groq in similarity accuracy when benchmarked against expert question banks.

- **OpenAI –**

  OpenAI's family of LLMs, including GPT-4, known for strong generative and reasoning capabilities.
  Works alongside Gemini in the follow-up question generation process, providing an additional candidate question for comparison and reinforcement.
  Justification: OpenAI's models were included to provide diversity in outputs and a benchmark against Gemini, ensuring the system selects the best-performing LLM for interrogation support.
- **Python**
  A high-level programming language widely used in AI research and development.
  Used to build a model evaluation framework for comparing LLMs, conducting

experiments on accuracy, and analyzing similarity percentages against expert question banks.
Justification: Python was chosen over alternatives like R or Julia because of its extensive AI ecosystem (PyTorch, TensorFlow, Hugging Face) and suitability for integrating NLP and ML experiments.

- **SBERT Sentence Transformers**
  A transformer-based model for semantic sentence embeddings.
  Used to compute cosine similarity scores between LLM-generated questions and expert-created question banks, enabling quantitative evaluation of model accuracy.
  Justification: Chosen over simpler similarity methods (TF-IDF, bag-of-words) due to its state-of-the-art semantic understanding and proven reliability in text similarity tasks.

- **Visual Studio Code – for development**
  A lightweight but powerful integrated development environment (IDE).
  The main development environment for coding, debugging, and integrating frontend, backend, and Python scripts.
  Justification: Preferred over JetBrains IDEs or Sublime Text due to its rich extension marketplace, GitHub integration, and strong TypeScript and Python support.
- **GitHub – Version Control**
  A platform for version control and collaborative software development using Git.
  Manages code versions, enables branching for new features, and integrates CI/CD pipelines for testing and deployment.
  Justification: GitHub was chosen over GitLab or Bitbucket because of its widespread adoption, seamless integration with VS Code, and collaboration features supporting a distributed research team.
- **Web Speech API**
  A browser-based API that enables speech recognition and synthesis. Used to capture suspect responses during interrogations, transcribing spoken answers into text that can be processed by the system.
  Justification: Chosen over third-party transcription services (Google Speech-to-Text) because it is lightweight, cost-effective, and natively supported by modern browsers, ensuring real-time responsiveness without reliance on external services.

2.6.System Requirements

Functional Requirements

The developed system incorporates a set of functional requirements that were designed, implemented, and validated to meet the practical needs of investigators. These requirements ensure that the solution is not only technically robust but also aligned with the operational context of real-world interrogations. The implemented functionalities are described below:

1. Real-time Response Analysis
   The system actively analyzes suspect responses during interrogation sessions in real time. Verbal inputs are transcribed into text using the Web Speech API, after which the responses are evaluated by LLMs The analysis identifies coherence, contextual relevance, and contradictions within the testimony. This functionality reduces delays associated with manual review and enhances the accuracy of detecting inconsistencies during questioning.

2. Dynamic Follow-up Question Generation
   Based on the analysis of responses, the system generates contextually relevant follow-up questions. Using prompts enriched with case data from police reports, the system queries to produce candidate questions. These are specifically designed to probe gaps, contradictions, or ambiguities, ensuring interrogations adapt to the evolving dialogue rather than relying on static, predefined question sets.

3. Investigator Review and Control
   To maintain human oversight, the system enables investigators to review, approve, reject, or modify generated follow-up questions before they are presented during interrogation. This ensures that AI suggestions supplement rather than replace investigator judgment, maintaining ethical compliance and procedural fairness while building investigator trust in the tool.

4. Feedback-driven Model Refinement
   The system integrates an investigator feedback mechanism that allows users to rate or adjust AI-generated questions. This feedback is logged and utilized in reinforcement learning loops to refine model performance over time. Positive feedback increases the likelihood of generating similar high-quality questions, while negative feedback suppresses less effective patterns.

5. Session Logging and Historical Records
   All interrogation data ,including generated questions, suspect responses, system analyses, and investigator feedback ,is stored securely in MongoDB. This maintains a comprehensive history of interrogation sessions, enabling investigators to review past interactions, identify patterns, and support case correlation. The log also provides an audit trail for transparency and accountability.
6. Integration with Other Investigative Modules
   The system has been designed to interoperate with additional components such as emotion analysis and incident correlation modules. For example, emotional cues derived from speech sentiment analysis or facial recognition systems can be integrated into the interrogation pipeline to enrich the context for question generation and response evaluation.
7. Multilingual Support
   Recognizing the diversity of linguistic contexts in Sri Lanka and beyond, the system supports multilingual interrogation sessions. The LLM models can process and generate questions in multiple languages, ensuring accessibility and effectiveness across varied linguistic backgrounds of victims, suspects, and witnesses.
8. Automated Reporting
   At the end of each session, the system automatically generates detailed reports summarizing interrogation progress. These reports include the sequence of questions, responses, detected inconsistencies, investigator feedback, and overall system performance. Reports can be exported for use in official case documentation and serve as valuable resources for case reviews and judicial processes.

Non-Functional Requirements

The non-functional requirements of the system were critical in ensuring that the implemented solution was not only technically sound but also reliable, secure, and practically usable in real investigative environments. The implemented NFRs are described below:

1. Low Latency and Real-Time Operation
   The system processes and analyzes suspect responses with a latency of under 2 seconds, enabling real-time interrogation support. Optimizations such as prompt engineering, efficient API routing in Next.js, and parallel model queries (Gemini

and GPT-4) ensure rapid response times. This responsiveness is essential for maintaining the natural flow of interrogations, preventing delays that could disrupt investigative continuity.

2. Scalability for Multi-Session Interrogations
The architecture was designed to scale efficiently across multiple simultaneous sessions. A microservices-based backend with containerization allows the system to allocate resources dynamically. This ensures that performance is not degraded when handling concurrent interrogations, making the solution suitable for deployment in larger law enforcement environments.

3. Responsive and Intuitive User Interface
The investigator-facing interface was developed using React, Next.js, Tailwind CSS, and ShadCN/UI components to provide a clean and consistent experience. Features such as real-time dashboards, session logs, and visual cues for inconsistencies enable investigators to interact with data seamlessly. This design promotes ease of adoption by officers with varying levels of technical expertise.

4. High Availability and Reliability
The system has been implemented to achieve 99.9% availability, ensuring minimal downtime for critical interrogation operations. Cloud-based hosting on Google Cloud Platform (GCP) with redundancy mechanisms ensures resilience against server outages. Automated backups and failover strategies further enhance system reliability.

5. Data Privacy and Security Compliance
Given the sensitivity of interrogation data, the system was built with strict adherence to data protection standards. Measures include encryption at rest and in transit (HTTPS/TLS), role-based authentication, anonymization of case data, and compliance with ethical/legal guidelines. This ensures the confidentiality of both victim and suspect testimonies while preventing unauthorized access or misuse.

6. Accessibility for Investigators
The user interface was developed with accessibility guidelines (WCAG 2.1 principles) in mind. Features include clear color contrasts, keyboard navigation, and screen-reader compatibility to ensure usability by investigators with disabilities. This adherence guarantees inclusivity in real-world operational contexts.

7. Cross-Platform Compatibility
The system supports multiple device environments, including desktop, mobile,

and tablet platforms. By leveraging responsive design frameworks within Tailwind CSS and React, investigators can access the system securely from different devices without loss of functionality, enabling flexibility in field and office usage.

8. Lightweight and Optimized Architecture
   The system employs a lightweight, modular architecture to minimize resource consumption while sustaining high performance. Model optimization techniques such as quantization and pruning were applied to reduce computational load, and caching strategies were integrated for faster repeated queries. This ensures that even with limited hardware resources, the system remains performant and scalable.

User Requirements

The system was designed with a user-centered approach, ensuring that investigators' practical needs and workflows were prioritized during implementation. The following user requirements were successfully implemented and validated:

1. Secure System Access
   Investigators are required to log in securely using their credentials before accessing the system. Authentication mechanisms were implemented using role-based access control and encrypted credentials, ensuring that only authorized personnel can utilize the question generation and response analysis features. This protects sensitive interrogation data from unauthorized access.

2. Real-Time Response Analysis Dashboard
   Investigators can view and review the AI's real-time analysis of suspect responses through an interactive dashboard. The dashboard presents detected inconsistencies, highlighted data points, missing information, and contextual insights derived from suspect testimonies. This ensures that investigators have a structured, evidence-driven view of the ongoing interrogation without the need for manual data processing.

3. Control Over AI-Generated Questions
   To maintain full investigator authority, the system allows users to approve, reject, or modify system-generated follow-up questions before they are deployed in interrogation. This feature ensures that AI acts as a supportive tool rather than a replacement for human judgment, maintaining procedural fairness and ethical standards.

4. Feedback Mechanism for Continuous Improvement
   Investigators can rate the relevance and effectiveness of AI-generated questions, feeding into a reinforcement learning loop. Positive feedback reinforces effective patterns in question generation, while negative feedback helps the system suppress less useful outputs. This continuous refinement mechanism ensures that the system evolves with practical use over time.

5. Access to Historical Session Data
   The system maintains a comprehensive session history, including generated questions, suspect responses, and investigator feedback. Investigators can review this historical data to analyze interrogation progress, track inconsistencies across multiple sessions, and identify patterns or behavioral trends. This log also provides an audit trail for transparency and accountability.

6. Automated Report Generation
   Investigators can generate and download detailed reports summarizing the interrogation session. These reports include AI analyses, highlighted contradictions, investigator-selected questions, and system performance metrics. The reports are formatted for integration into official case documentation, supporting judicial and investigative needs.

7. Multilingual Interface Support
   The system provides an option for investigators to switch the interface language to accommodate multilingual usage. This feature enables question generation and response analysis in multiple languages, ensuring that interrogations can be conducted effectively in diverse linguistic contexts, such as Sinhala, Tamil, and English in Sri Lanka.

8. Real-Time Awareness of Inconsistencies
   The dashboard is investigators in real time when inconsistencies, contradictions, or missing information are detected in suspect responses. This ensures that investigators are immediately aware of critical issues and can act on them without delay, strengthening the interrogation process.

2.7. Domain Specific Considerations

Security Considerations

Given the sensitivity of interrogation data and the high stakes involved in criminal investigations, security is one of the most critical domain-specific considerations in the design and deployment of the Dynamic Question Generation and Response Analysis system. Protecting suspect testimonies, case records, and investigator decisions from unauthorized access or misuse is vital for maintaining the integrity of the legal process and ensuring trust in AI-assisted tools. The implemented system integrates multiple layers of security measures to address confidentiality, integrity, availability, and compliance.

1. Data Confidentiality

The system handles sensitive interrogation data, including testimonies, generated questions, and response analyses, which must be kept confidential. To achieve this:

- Encryption at rest and in transit is enforced using industry-standard protocols
- Anonymization pipelines strip all personally identifiable information (PII) from police reports and testimonies before they are processed by Large Language Models (LLMs), ensuring that the data used for training or analysis cannot be traced back to individuals.
- Role-based access control (RBAC) restricts system access to authorized investigators only, with audit logs maintained for every login and session activity.

2. Data Integrity

Maintaining the accuracy and authenticity of interrogation data is essential to ensure that evidence is admissible and tamper-proof. To safeguard integrity:

- Immutable logging in MongoDB ensures that all interrogation records, including Q&A pairs and feedback, are securely timestamped and protected against unauthorized modifications.
- Digital audit trails allow supervisors and judicial authorities to verify that AI-generated outputs were not altered after the session, providing transparency and accountability.
- Version control for AI models and prompts ensures that every interrogation can be traced back to the specific model configuration used, preventing disputes over system reliability.

3. System Availability and Reliability

Downtime in critical operations such as interrogations could undermine investigations. To mitigate this risk:

- The system is hosted on Google Cloud Platform (GCP) with redundancy and failover mechanisms to ensure 99.9% availability.

- Automated backup and recovery protocols preserve interrogation logs and prevent data loss in case of hardware or system failures.

4. Access Security

Since investigators interact directly with the system during live interrogations, robust access security is required:

- Multi-factor authentication (MFA) strengthens login security and prevents unauthorized entry even if credentials are compromised.
- Session timeouts and automatic logouts prevent accidental exposure of sensitive data when workstations are left unattended.
- Fine-grained permissions ensure that investigators, supervisors, and administrators have different levels of access, protecting critical backend functions.

5. Protection Against AI-Specific Threats

Given the reliance on LLMs and NLP models, additional risks must be addressed:

- Prompt injection attacks are mitigated by sanitizing input data before passing it to Gemini or GPT, preventing malicious manipulation of AI outputs.
- Bias monitoring and mitigation mechanisms are in place to prevent systemic discrimination in AI-generated questions, ensuring fairness across cases.
- Model isolation ensures that training data and interrogation inputs are not exposed to external services beyond the secured APIs.

6. Legal and Ethical Compliance

Finally, the system's security framework is closely tied to legal and ethical compliance:

- Adheres to Sri Lanka's Code of Criminal Procedure Act and aligns with international data protection standards such as GDPR for handling sensitive data.
- Security policies ensure that data access and usage comply with judicial admissibility standards, meaning AI outputs can withstand legal scrutiny.
- Ethical safeguards, such as investigator oversight and prevention of coercive questioning, are embedded to prevent abuse of the system.

Social Considerations

The deployment of an AI-assisted interrogation system in criminal investigations extends beyond technical and legal dimensions; it has profound social implications. Since the system deals with sensitive testimonies, victims of trauma, and accused individuals under investigation, its adoption must be carefully examined in relation to public trust, investigatory practices, and the broader justice system. The following social considerations were considered in the design and implementation of the system:

1. Public Trust and Perception

For AI systems in law enforcement to be accepted, they must inspire confidence among the public. There is often skepticism toward automated decision-making in sensitive areas such as criminal justice. To address this:

- The system was designed to act as a support tool rather than a replacement for investigators, ensuring that ultimate responsibility remains with human officers.
- Transparency measures, such as clear audit trails and investigator oversight of AI-generated questions, help build public confidence in fairness and accountability.
- By reducing investigator fatigue and human error, the system strengthens the credibility of testimonies and outcomes, fostering greater trust in legal processes.

2. Victim Sensitivity and Ethical Interactions

Victims of crime, particularly in cases involving violence or abuse, may already be traumatized during testimony. Technology must not exacerbate their experience:

- The system generates contextually relevant but non-coercive follow-up questions, reducing the likelihood of repetitive or insensitive questioning.
- Emotional and behavioral cues captured through multimodal analysis provide investigators with insights into victim stress levels, allowing for a more empathetic approach.
- This helps ensure that the system contributes positively to victim-centered justice, an essential principle in modern investigative practice.

3. Investigator Adaptation and Training

Introducing AI into established policing practices requires cultural and behavioral change within law enforcement agencies:

- Investigators must be trained to understand how the system generates suggestions, interpret its outputs, and integrate them responsibly into their workflows.
- The system interface was designed with usability and accessibility in mind, making it approachable even for personnel with limited technical expertise.
- Proper training and gradual adoption will ensure that the technology is perceived as a practical aid rather than an intrusive burden, reducing resistance from practitioners.

4. Social Equity and Bias Concerns

AI systems risk perpetuating or amplifying social inequalities if not designed with fairness in mind:

- The system's multilingual capabilities allow interrogation support in Sinhala, Tamil, and English, ensuring inclusivity across Sri Lanka's diverse population.
- Bias monitoring mechanisms were integrated to minimize the risk of discriminatory questioning patterns that could unfairly target individuals based on language, ethnicity, or socio-economic status.
- By focusing on data anonymization and neutral question framing, the system supports equitable treatment across different social groups, reinforcing the principle of justice for all.

5. Wider Societal Impact

Beyond individual interrogations, the system contributes to broader societal goals:

- By improving interrogation efficiency and accuracy, it helps accelerate case resolution, reducing the backlog of unresolved cases in the justice system.
- The system indirectly strengthens public safety and community well-being by enabling investigators to uncover critical links between incidents more effectively.
- Responsible deployment sets a precedent for the ethical use of AI in governance, paving the way for more socially responsible adoption of advanced technologies in Sri Lanka and beyond.

Ethical Considerations

The integration of AI into criminal investigations raises critical ethical challenges, particularly when applied in sensitive contexts such as interrogations. Unlike purely technical domains, law enforcement operates within a framework where human rights, fairness, and accountability must be preserved at all times. The following ethical considerations guided the design and implementation of the Dynamic Question Generation and Response Analysis system:

1. Human Oversight and Accountability

While the system automates response analysis and question generation, ethical safeguards ensure that final authority rests with investigators. The system presents AI-generated questions as recommendations only, requiring explicit approval, rejection, or modification by the investigator before use. This design prevents overreliance on AI, ensuring that accountability for interrogation practices remains human-centered and legally sound.

2. Prevention of Coercion

Interrogations inherently involve power imbalances, which may be amplified if AI-generated questions are misused. To avoid coercive or manipulative questioning:

- The system is programmed to generate contextually relevant but neutral questions, avoiding language that could pressure or intimidate suspects or victims.
- Feedback mechanisms allow investigators to flag inappropriate or biased questions, improving future system outputs.
- The ethical principle of informed consent is respected, as suspects and witnesses must remain aware that interrogation decisions are ultimately investigator-driven, not dictated by an algorithm.

3. Fairness and Bias Mitigation

AI models can inadvertently introduce biases inherited from training data. In criminal justice, such biases could have severe consequences, leading to unequal treatment of individuals based on race, language, gender, or socio-economic status. To mitigate this:

- Training datasets undergo anonymization and preprocessing to remove potentially discriminatory features.
- The system incorporates bias detection and monitoring tools to identify unfair questioning patterns.

- Multilingual support (Sinhala, Tamil, English) ensures that individuals are interrogated in their native or preferred language, reducing linguistic discrimination.

4. Privacy and Confidentiality

Ethical handling of interrogation data is crucial for protecting the rights of suspects, victims, and witnesses:

- Personally identifiable information (PII) is anonymized before being processed by LLMs, ensuring compliance with privacy laws.
- Strict data retention policies are enforced so that interrogation data is not stored indefinitely, reducing risks of misuse.
- Confidentiality safeguards prevent unauthorized access to sensitive case information, maintaining the dignity and rights of those involved.

5. Transparency and Explainability

For AI systems to be ethically defensible in judicial contexts, their outputs must be transparent and explainable:

- Every generated question and system analysis is logged with a traceable audit trail, allowing investigators, supervisors, or courts to review the reasoning process.
- The system provides clear summaries of why certain inconsistencies or contradictions were flagged, avoiding “black box” decision-making.
- This ensures that the system can withstand judicial scrutiny and that its contributions are verifiable and trustworthy.

6. Ethical Use of AI in Law Enforcement

The project acknowledges the broader ethical debate surrounding AI in policing. To avoid misuse:

- The system is explicitly designed as a support tool, not as a decision-maker in determining guilt or innocence.

- Ethical guidelines prevent integration with surveillance technologies that could infringe on civil liberties.
- Continuous consultation with law enforcement professionals and an external retired police supervisor ensures that the system is aligned with ethical policing practices in Sri Lanka's context.

7. Long-Term Ethical Responsibility

Finally, ethical responsibility extends beyond deployment:

- Regular ethical audits are recommended to monitor system performance, fairness, and compliance with evolving legal standards.
- Investigators are provided with training on ethical system use, reinforcing human rights considerations in every interrogation.
- The system promotes a culture of responsible AI adoption, setting a benchmark for other AI initiatives in law enforcement.

3. Testing & Results Discussion

The Dynamic Question Generation and Response Analysis component was designed to generate investigative questions from structured case data, analyze responses for relevance and consistency, and iteratively propose follow-up questions. The methodology integrates natural language processing (NLP), large language models (LLMs), and semantic similarity scoring to ensure generated questions meet the standards of human-curated investigative question banks. The evaluation framework compared three LLMs, Gemini, OpenAI, and Groq ,to determine the most effective models for question generation in the context of criminal investigations.

Structured case attributes were retrieved from MongoDB collections that included incident metadata, victim and suspect information, evidence mentions, and police case details. From this data, initial questions were generated by LLMs using templated prompts aligned to investigative practices.

Each LLM, Gemini, OpenAI, and Groq, was tasked with generating investigative questions for the same structured inputs. The outputs were standardized by removing duplicates and enforcing grammatical correctness to ensure fair comparison across models. The generated questions were then evaluated against the human question bank of 98 expert-curated investigative questions that served as the gold standard for evaluation.

The evaluation was conducted separately for Gemini, OpenAI, and Groq across identical input conditions. Scores were computed for each model using the similarity pipeline.

1. **Embedding Computation**: Both generated and human reference questions were encoded using Sentence-BERT (*all-MiniLM-L6-v2*), producing dense semantic embeddings.
2. **Cosine Similarity Matrix**: A pairwise cosine similarity matrix was computed between generated and reference embeddings.
3. **Hungarian Assignment**: The Hungarian algorithm was applied to maximize one-to-one matching across the similarity matrix. This prevented multiple generated questions from being aligned to the same reference question and eliminated double-counting.
4. **Similarity Thresholding**: A match was considered valid if the cosine similarity exceeded the canonical threshold $\tau = 0.70$.

5. **Final Score Calculation**: The Similarity Percentage was defined as:

Similarity %=#matched pairs with cosine ≥ τ98×100\text{Similarity \%} = \frac{\#\text{matched pairs with cosine ≥ τ}}{98} \times 100Similarity %=98#matched pairs with cosine ≥ τ×100

This scoring framework (SBERT + cosine + Hungarian @ τ=0.70) was established as the project's canonical metric and has been fully implemented in both code and documentation.

While semantic similarity was the primary performance indicator, additional factors such as process time and cost were incorporated into the evaluation to reflect practical deployment considerations. Each factor was assigned a weight:

- Similarity Score → 60%
- Process Time → 10%
- Cost → 30%

The evaluation results are shown in Table.

Model	**Similarity Score (%)**	**Process Time (s)**	**Cost**	**Weighted Score**
Gemini	**86.73**	**13.26**	**Free**	**81.2**
OpenAI	**78.57**	**13.66**	**Pay-As-You-Go**	**58.3**
Groq	**36.73**	**4.56**	**Free**	**26.8**

Table 1 : Evaluation Table for Comparison of LLMs

(Weighted Score = 0.6 × Similarity + 0.1 × (scaled inverse process time) + 0.3 × Cost Score, where cost advantage is scaled as Free = 100, Pay-As-You-Go = 0.)

Based on the weighted evaluation, **Gemini achieved the highest overall score (81.2)**, making it the most effective LLM for dynamic question generation. Gemini outperformed both OpenAI and Groq by balancing high similarity with zero cost, despite slightly higher processing time compared to Groq. OpenAI demonstrated competitive similarity but was penalized due to cost, while Groq's low semantic accuracy made it unsuitable despite its speed advantage.

4. Commercialization

Market Gap

The criminal justice system currently struggles with inefficiencies caused by manual handling of investigation data, leading to delays, errors, and overlooked critical correlations among cases. Existing tools are mostly standalone solutions addressing narrow tasks such as lie detection or basic database management, lacking an integrated AI-powered platform that combines automated data extraction, emotional and behavioral analysis, dynamic interrogation support, and multi-modal incident correlation.

A clear gap exists for a comprehensive, AI-driven investigative assistance system that dramatically improves investigative accuracy, reduces case resolution time, and enhances decision-making through real-time insights and integrations with law enforcement databases.

Value Proposition

- End-to-End AI Integration: Combines OCR, NLP, multi-modal emotion analysis, and pattern recognition to deliver a unified investigative tool.
- Efficiency Gains: Significantly reduces manual data handling, accelerating case resolution and improving investigator productivity.
- Enhanced Accuracy: Advanced LLM and multimodal AI models ensure high precision in data extraction and behavioral insights.
- Real-Time Decision Support: Dynamic questioning and emotion tracking assist interrogators in eliciting truthful and relevant information.
- Seamless Data Synchronization: Integrates with law enforcement databases to maintain updated and centralized case data.
- User-Friendly Interface: Intuitive dashboards with gamification features encourage user engagement and ease of adoption.
- Multi-Lingual and Accessibility Support: Cater to diverse user bases while complying with accessibility standards.

Target Audience

- Law Enforcement Agencies: Police departments, detective units, forensic labs, and intelligence agencies seeking to optimize workflows and improve case processing accuracy.
- Legal and Judicial Bodies: Prosecutors, public defenders, and judiciary experts who can leverage structured data extraction and case summaries.
- Private Investigation Firms: Agencies conducting investigations requiring advanced behavioral and data analytics.
- Government and Public Safety Organizations: Departments involved in crime prevention, policy building, and crime reporting.
- Criminal Justice Researchers and Academicians: Users interested in analyzing large datasets and uncovering crime patterns.
- Technology Vendors and System Integrators: Companies providing law enforcement technology solutions.

Commercialization Strategy

- Pilot Projects and Collaborations: Partner with selected police departments and forensic labs to deploy pilot versions, gather real-world feedback, and demonstrate value.
- Government and Institutional Procurement: Participate in tenders and public sector technology initiatives for law enforcement modernization.
- SaaS and Licensing Model: Offer a subscription-based cloud service with tiered pricing depending on usage, user seats, and feature access.
- Training and Certification Programs: Develop comprehensive training modules and certification tracks for investigative personnel to ensure effective adoption.
- Technology Partnerships: Collaborate with established law enforcement tech vendors for integration and co-marketing opportunities.
- Conferences and Workshops: Present the solution at criminal justice and AI conferences to build reputation and acquire leads.
- Online Presence and Thought Leadership: Maintain a strong digital presence through content marketing, webinars, whitepapers, and case studies.

- Support and Customer Success: Provide robust post-deployment support via dedicated teams to enhance customer satisfaction and retention.

Revenue Models

- Subscription Fees: Monthly or yearly SaaS subscriptions offering variable features based on plan levels (basic, professional, enterprise).
- Implementation and Customization Charges: Fees for initial setup, customization to agency workflows, and integration services.
- Training and Certification Fees: Revenue from training programs and accredited investigator certification.
- Consulting Services: Expert consulting for data management strategies, AI adoption planning, and forensic analytics optimization.
- Data Analytics Add-ons: Premium modules for advanced analytics, automated reporting, and AI model tuning.
- Government Grants and Funding: Potential funding support from public safety innovation grants.

Community Building

- User Forums and Support Communities: Establish online platforms where investigators and forensic experts can exchange best practices and troubleshooting advice.
- Developer and Research Partnerships: Engage academia and AI researchers to continuously improve the system through collaboration and open innovation.
- Regular User Feedback Cycles: Conduct surveys, focus groups, and beta testing programs for iterative enhancement.
- Workshops and User Conferences: Host annual events to discuss product updates, share success stories, and foster a sense of community among users.
- Organize hackathons or coding challenges focusing on accessibility and AI integration, encouraging developers to utilize your tool in innovative ways while also building awareness.

Future Development

- Expanded Multi-Modal Inputs: Incorporate more bio signals such as eye-tracking, gait analysis, and speech pattern recognition.
- Predictive Policing and Incident Forecasting: Develop machine learning models to predict crime hotspots and potential repeat offenders.
- Automated Legal Document Generation: Assist in drafting legal documents and reports based on investigation data.
- Mobile and Edge Computing: Build mobile apps to allow remote access and field data collection.
- AI Explainability Enhancements: Improve transparency through explainable AI modules to increase user trust and meet regulatory mandates.
- Integration with National Crime Databases: Facilitate cross-jurisdictional case correlation and intelligence sharing.
- 19 User Feedback Loop: Establish a continuous feedback loop with users to identify trends and requests for new features. Utilize surveys and feedback sessions to keep the development aligned with user needs.

Risk Mitigation and Compliance

- Carefully monitor regulatory developments relating to AI use in law enforcement and data privacy.
- Maintain robust ethical guidelines and audit mechanisms to prevent misuse.
- Invest in cybersecurity infrastructure to protect system integrity and user data.

5. Conclusion

The developed AI-driven system for dynamic question generation and response analysis significantly advances the state of criminal investigations by addressing the drawbacks of manual interrogation methodologies. By integrating cutting-edge NLP and large language models with real-time response evaluation and behavioral analytics, the system provides investigators with adaptive, context-aware questioning capabilities that improve the depth, accuracy, and consistency of testimony analysis. The inclusion of investigator oversight ensures that AI serves as a supportive tool rather than a decision-maker, preserving accountability and ethical standards. Robust security, data privacy frameworks, and legal compliance measures safeguard sensitive interrogation data and promote judicial acceptability.

Empirical evaluation confirmed that Google's Gemini model offers the best balance of semantic relevance, processing efficiency, and cost-effectiveness for dynamic question generation. The multi-model approach, which also incorporates OpenAI's GPT-4, enriches question diversity and adaptability. The user-friendly interface, real-time dashboards, and feedback-driven learning loop enhance investigator usability and trust, enabling continuous system improvement and operational alignment through field feedback.

This solution addresses critical criminal justice system inefficiencies, notably by reducing investigator cognitive load, minimizing human errors, and accelerating interrogation workflows. The system's multilingual support and non-coercive questioning protocols foster inclusivity and fairness. By enabling thorough, evidence-based interviewing practices, the platform has the potential to expedite case resolutions and strengthen confidence in judicial outcomes. Future expansion opportunities include integrating additional multimodal inputs, predictive policing tools, mobile deployments, and national crime database connectivity. The project establishes a foundational AI-assisted framework that can be further refined and scaled, setting the stage for ethically responsible and technologically sophisticated criminal investigations in Sri Lanka and beyond.

6. References

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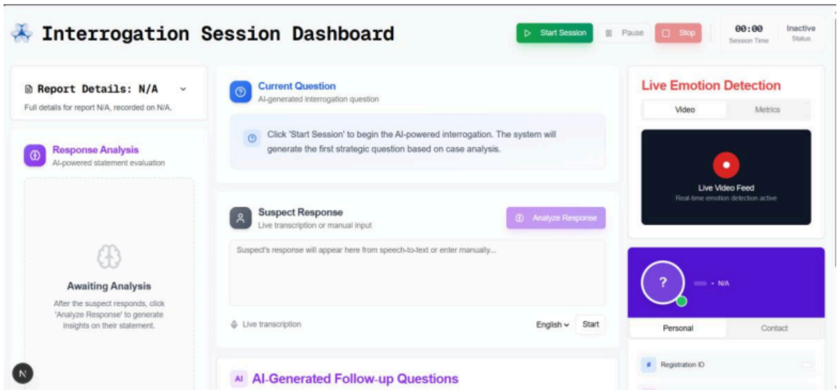
Appendix



Interrogation Session Dashboard
Start Session
Pause
Stop
00:00
Session Time
Inactive
Status
Report Details: N/A
Full details for report N/A, recorded on N/A.
Response Analysis
AI-powered statement evaluation
Awaiting Analysis
After the suspect responds, click 'Analyze Response' to generate insights on their statement.
Current Question
AI-generated interrogation question
Click 'Start Session' to begin the AI-powered interrogation. The system will generate the first strategic question based on case analysis.
Suspect Response
Live transcription or manual input
Analyze Response
Suspect's response will appear here from speech-to-text or enter manually...
Live transcription
English
Start
AI-Generated Follow-up Questions
Live Emotion Detection
Video
Metrics
Live Video Feed
Real-time emotion detection active
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Contact
Registration ID

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